People's Democratic Republic of Algeria
Ministry of Higher Education and Scientific Research

Laboratoire STIC

Mobile Operating Systems

Session 3: General Architecture of Mobile Operating Systems

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Opening Question

When you tap an app icon, what happens inside the OS?

- Does the app talk directly to hardware?
- Who decides memory allocation?
- Where is security enforced?

Session Objectives

By the end of this session, you will be able to:

- Understand the generic layered architecture of mobile OS
- Identify kernel, middleware, and application layers
- Compare mobile OS architecture with classical OS
- Explain why layering is essential for security and performance

Why Do We Need an Architecture?

Architecture answers:

- Who does what?
- Who can access what?
- How complexity is managed?
- How the system evolves safely?

Architecture as a Contract

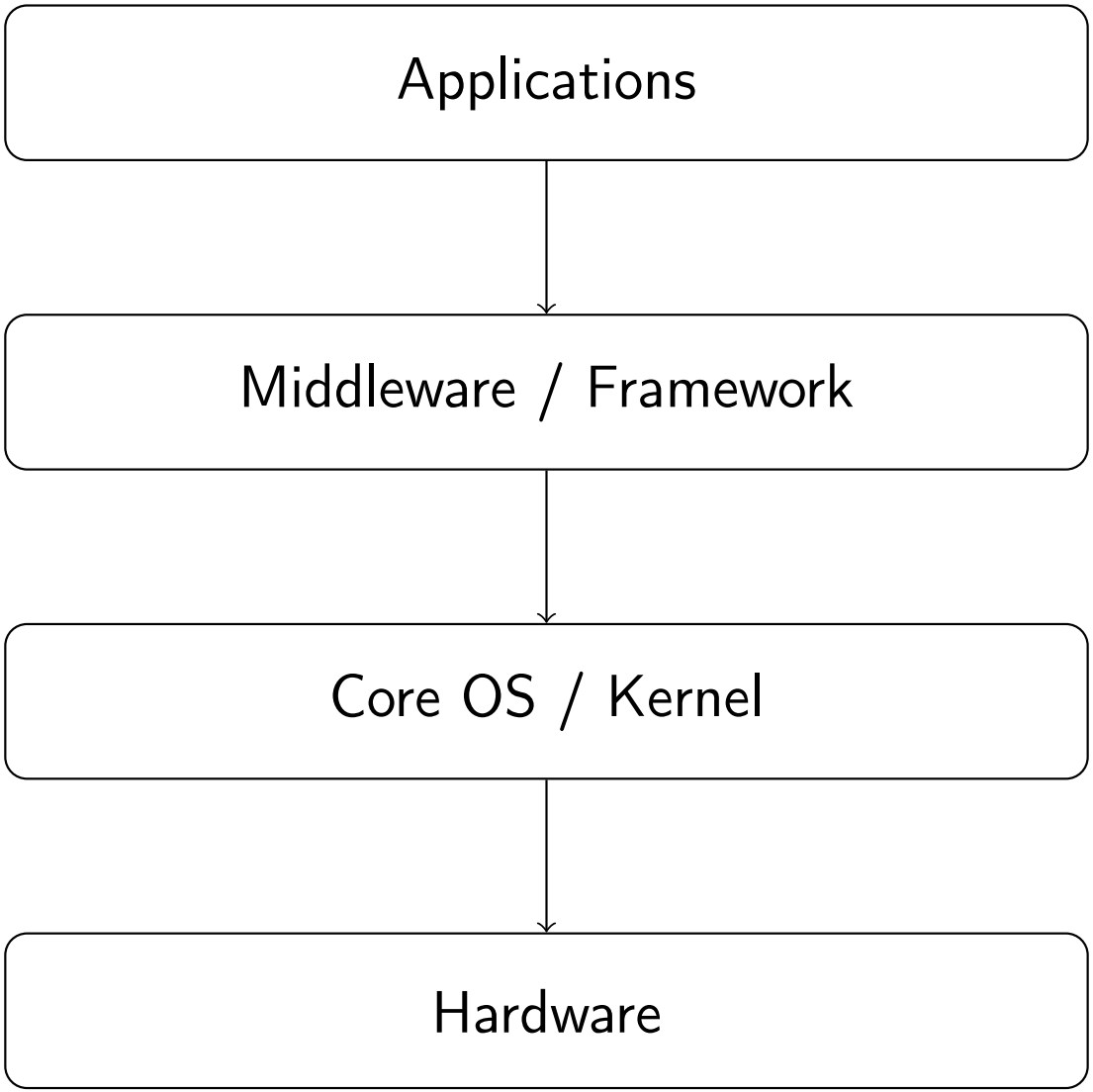
- Defines clear responsibilities
- Limits coupling between components
- Protects system integrity
- Enables parallel development

The Generic Mobile OS Architecture

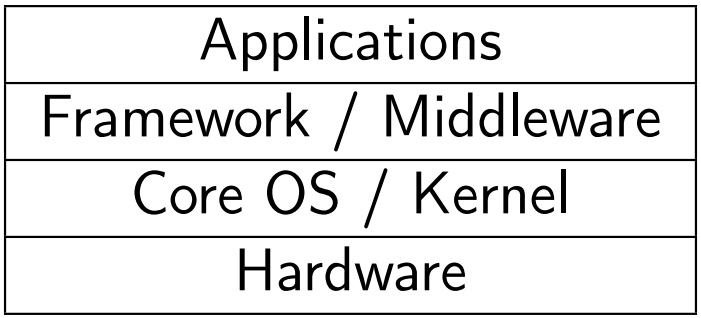
Most mobile OS follow a layered model:

1. Applications
2. Middleware / Framework
3. Core OS / Kernel

Generic Mobile OS Layered Architecture



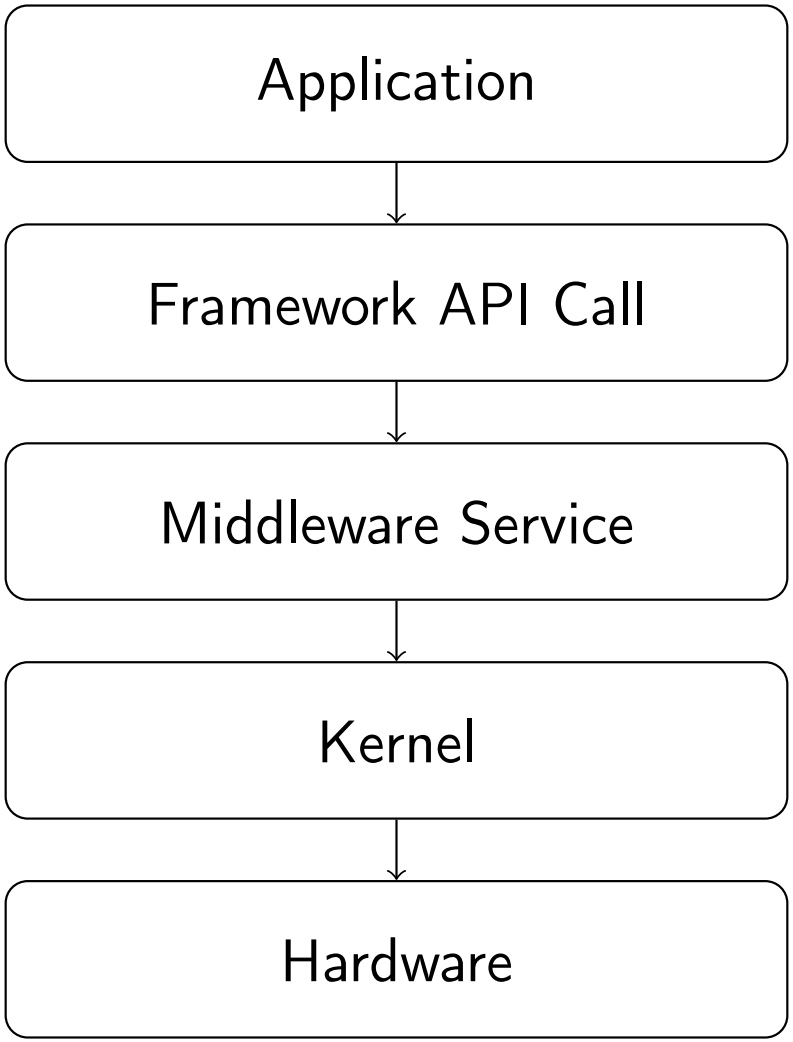
High-Level View



Application Layer — Role

- Contains user-facing applications
- No direct hardware access
- Runs in sandboxed environments
- Uses OS-provided APIs

How an Application Executes (Simplified Flow)



Why Applications Are Isolated

- Prevent malicious behavior
- Avoid crashes propagating
- Protect user data

Pedagogical Question

Why shouldn't apps access hardware directly?

Middleware — Definition

Middleware is the software layer that:

- Exposes system services to apps
- Hides hardware complexity
- Enforces security and policies

Typical Middleware Services

- UI toolkit
- Resource management
- Location services
- Media services
- Notification system

Middleware as a Policy Enforcer

- Checks permissions
- Controls background execution
- Regulates access to sensors

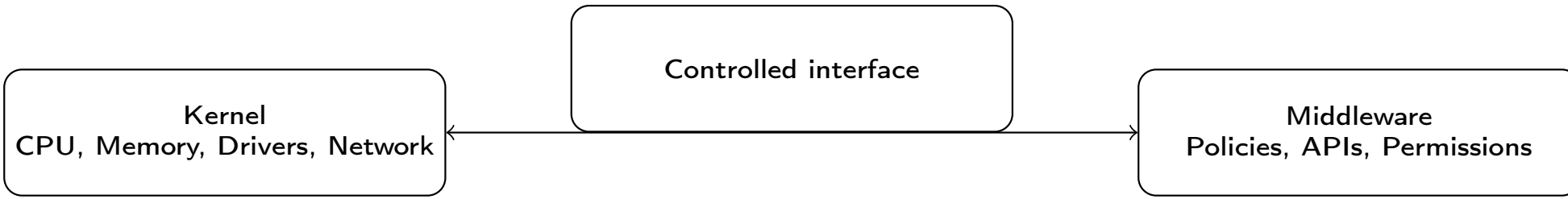
Kernel — Core Responsibilities

- Process management
- Memory management
- File systems
- Networking
- Device drivers

Kernel vs Middleware

- Kernel: low-level, hardware-near
- Middleware: policy and abstraction
- Apps never call kernel directly

Kernel vs Middleware — Conceptual Separation



Energy Management Location

- Kernel handles CPU power states
- Middleware decides when tasks run
- Apps must cooperate

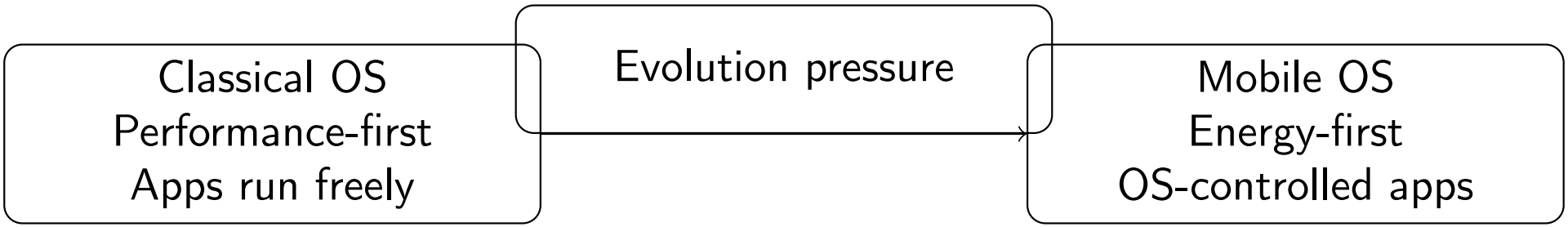
Classical OS Architecture (Reminder)

- Desktop/server OS focus on performance
- Apps often run continuously
- Less aggressive lifecycle control

Key Differences with Mobile OS

- Mobile OS: energy-first design
- Strong app lifecycle control
- Permission-based security
- Touch-centric UI

Classical OS vs Mobile OS — Architectural Emphasis



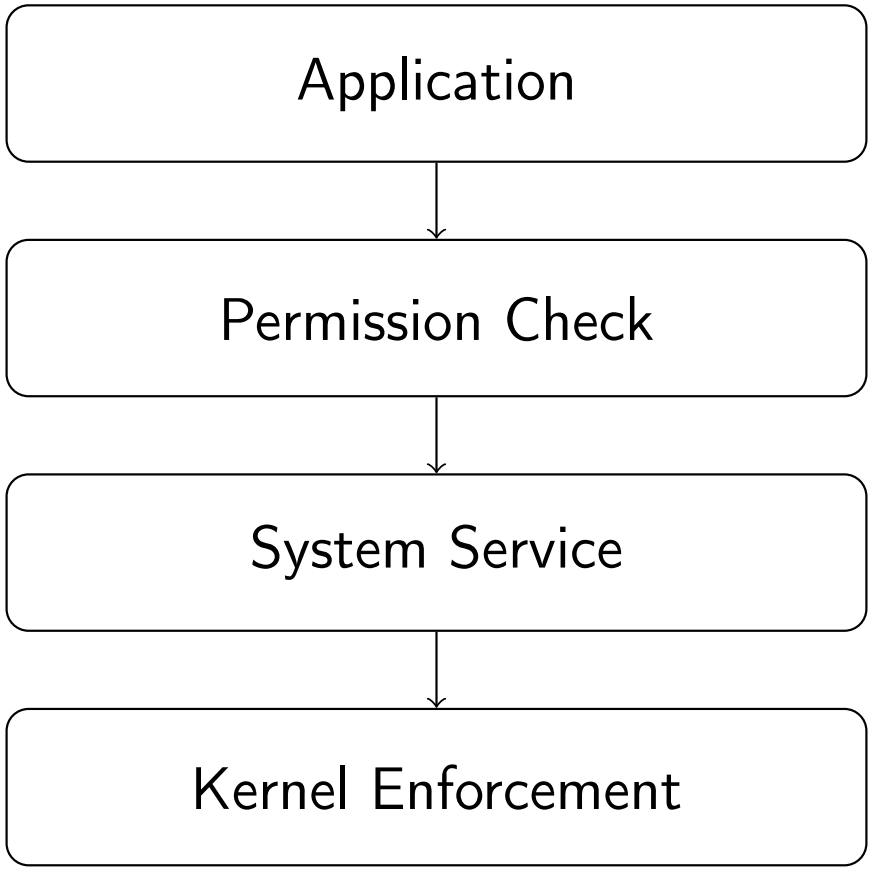
Comparison Table

Aspect	Classical OS	Mobile OS
Energy	Secondary	Primary
App lifetime	Long-running	OS-controlled
Security	User-based	Permission-based
Hardware diversity	Limited	Very high

Security Benefits

- Limits attack surface
- Prevents privilege escalation
- Enables permission enforcement

Security Enforcement Through Layers



Performance Benefits

- Optimized hardware access
- Centralized scheduling
- Reduced redundant work

Maintainability and Evolution

- Easier OS updates
- Backward compatibility
- Vendor customization

Activity: Layer Identification

Question: Where does each belong?

- Camera access API
- Process scheduler
- WhatsApp UI
- File system

Activity — Solution

- Camera API → Middleware
- Scheduler → Kernel
- WhatsApp UI → Application
- File system → Kernel

Key Takeaways

- Mobile OS use layered architecture
- Apps never access hardware directly
- Middleware enforces policy
- Kernel ensures efficiency and stability

Looking Ahead

Next session (Session 4): iOS Overview & History

We will:

- Apply this architecture to iOS
- Explore Apple's design philosophy
- Study multitasking in iOS

End of Session 3

Thank you — see you next Sunday